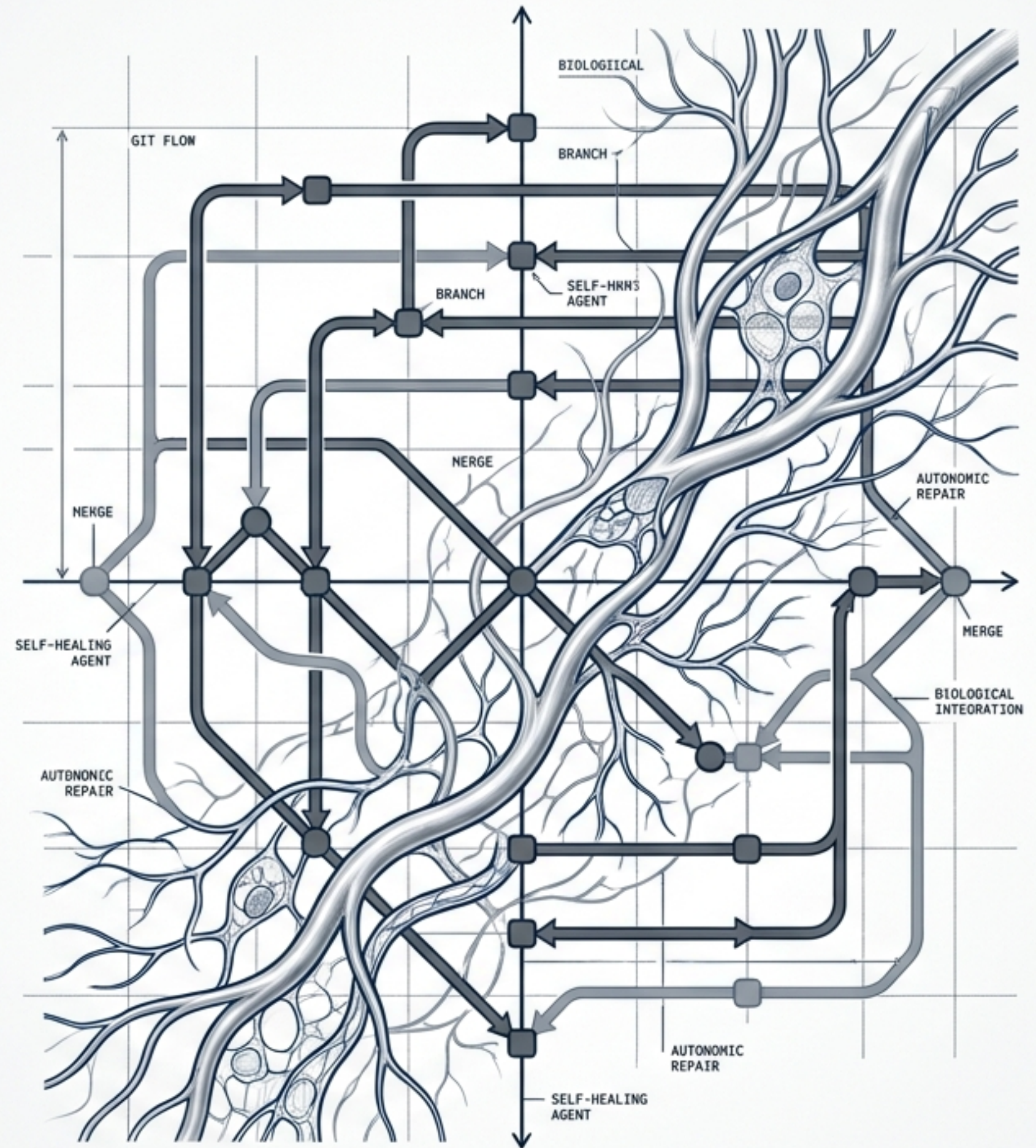
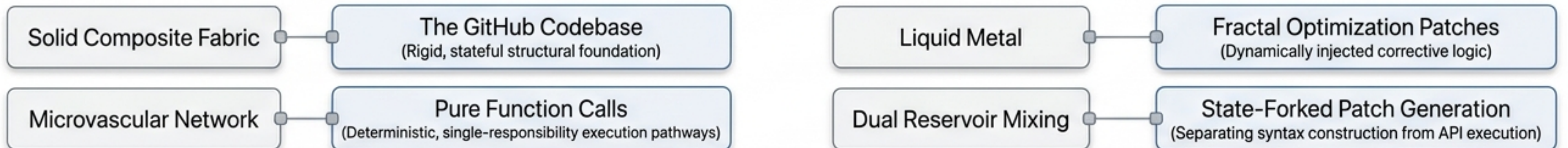
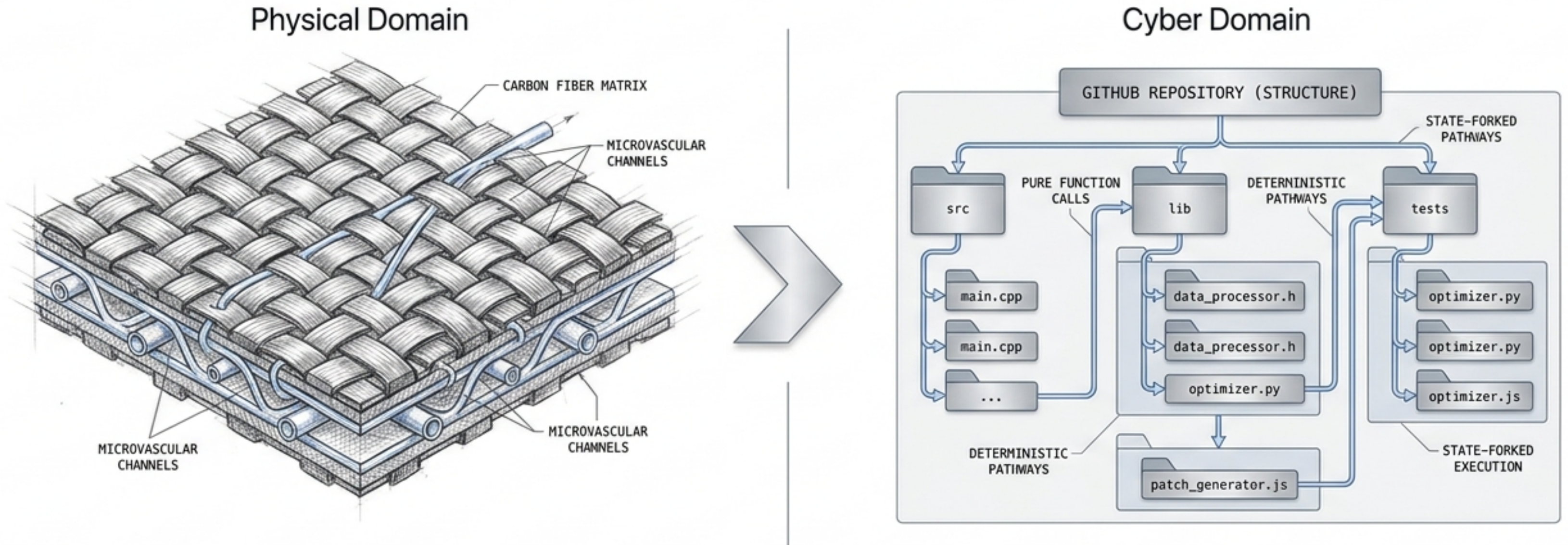


The VEINS Playbook: Architecting Immortal GitHub Agent Fleets

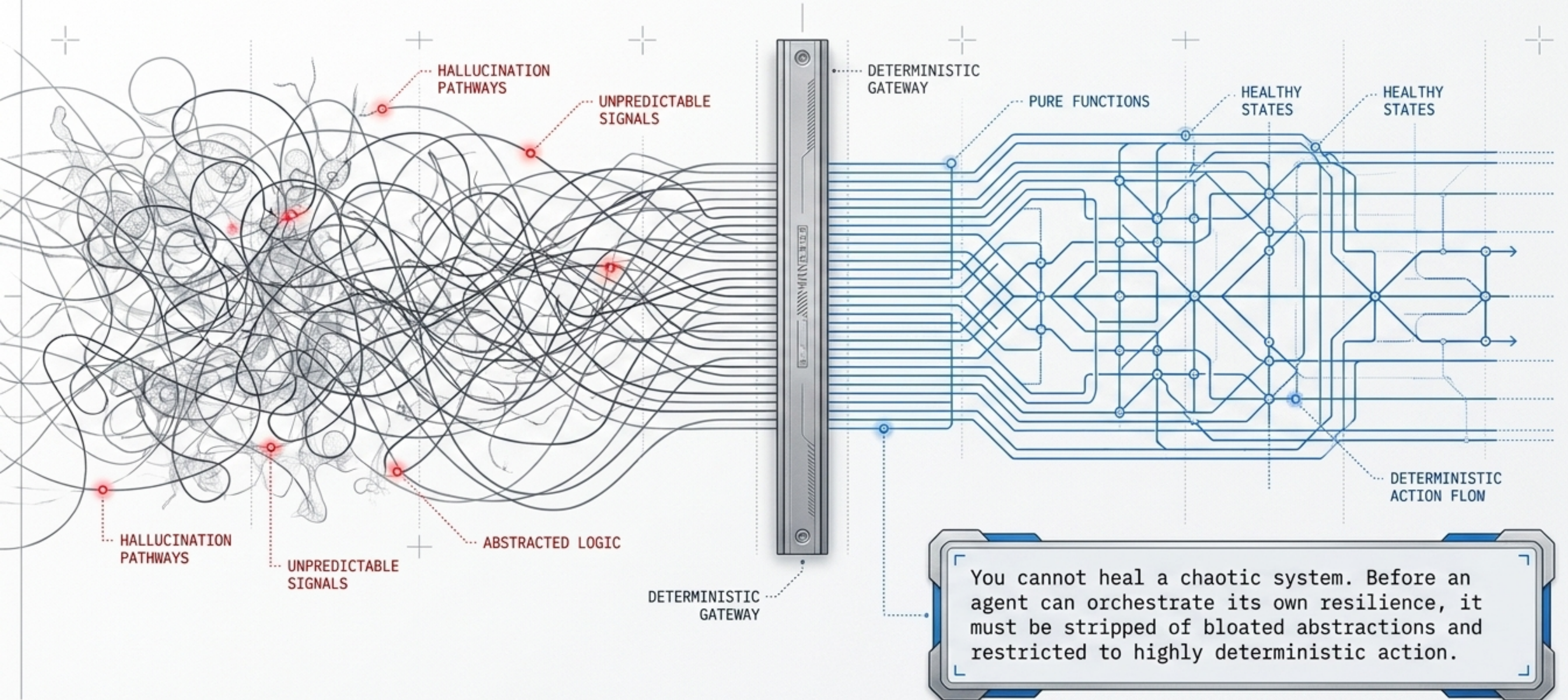
A definitive methodology for
self-healing, autonomic resilience
in production-grade AI systems.



Translating Physical Resilience to the Codebase



Phase 1: The Beginning (Deterministic Foundations)



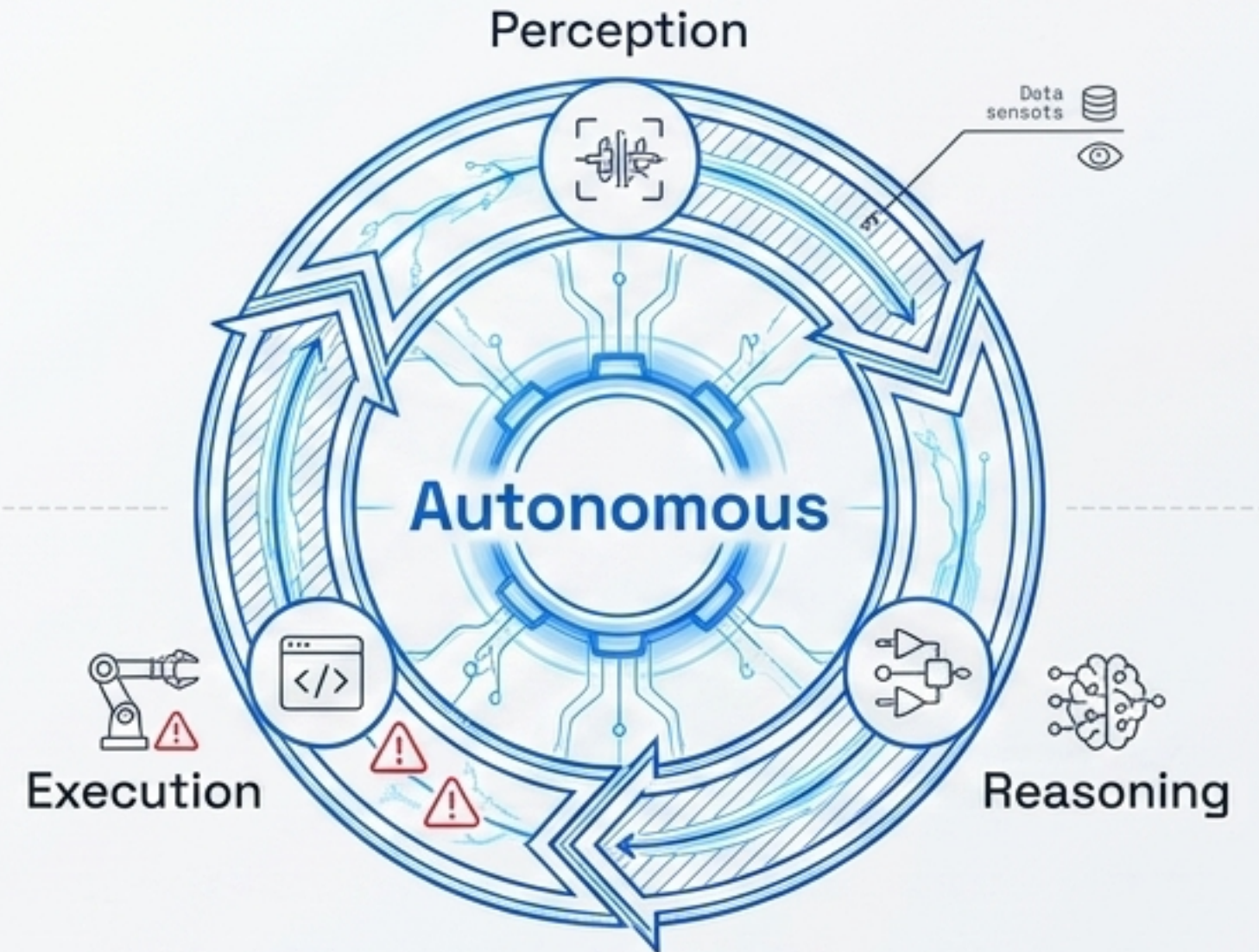
The Inversion Inversion of Control Flow

The Tennis Match (Traditional)



The LLM acts as sophisticated autocomplete.

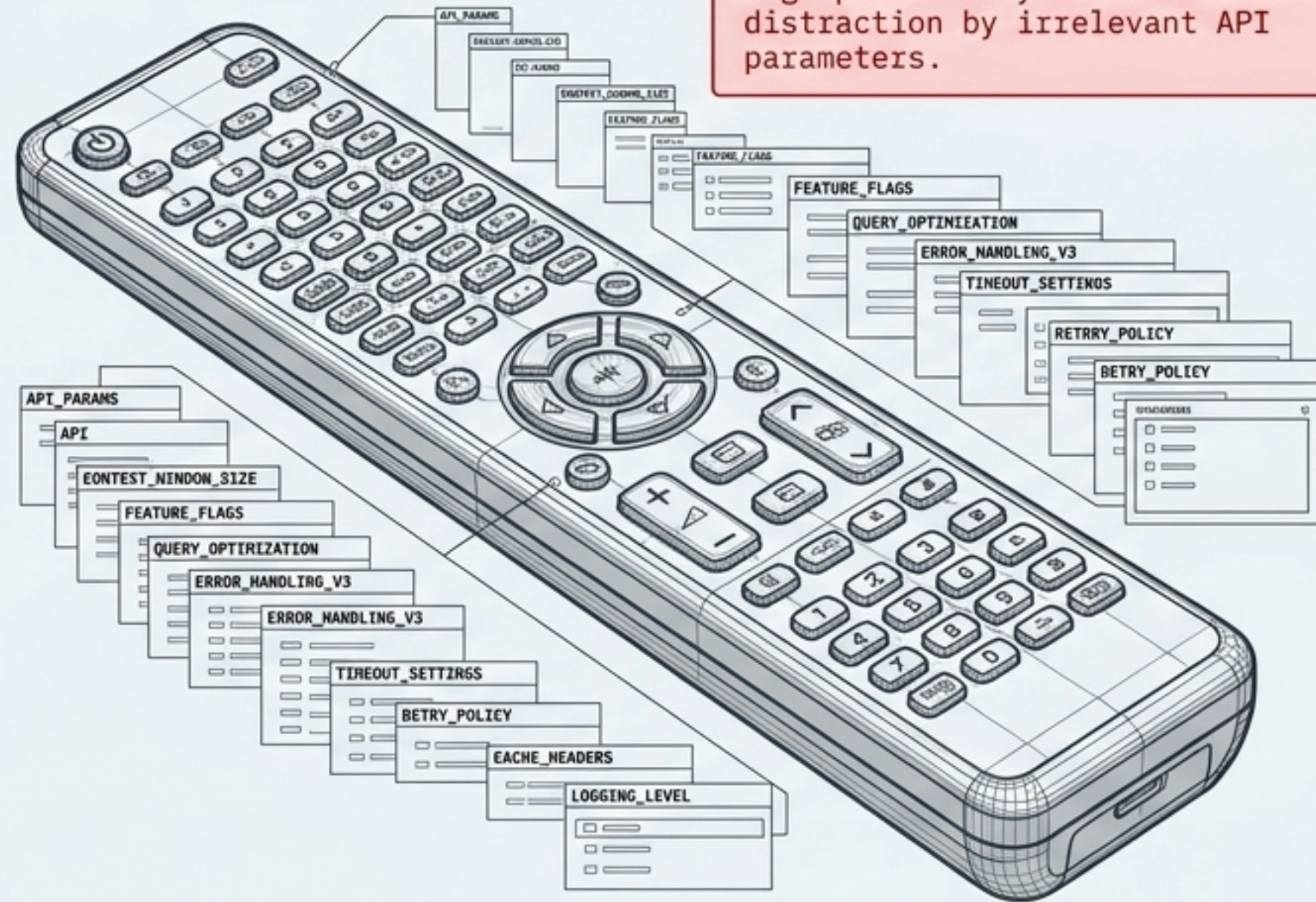
The Continuous Engine (Agentic)



The agent queries its environment, evaluates data, and triggers GitHub actions without human intervention.

The Threat of Cognitive Overload

MCP Server (Noise)



ATTENTION MECHANISM DILUTION:
High probability of mathematical distraction by irrelevant API parameters.

Pure Function (Signal)

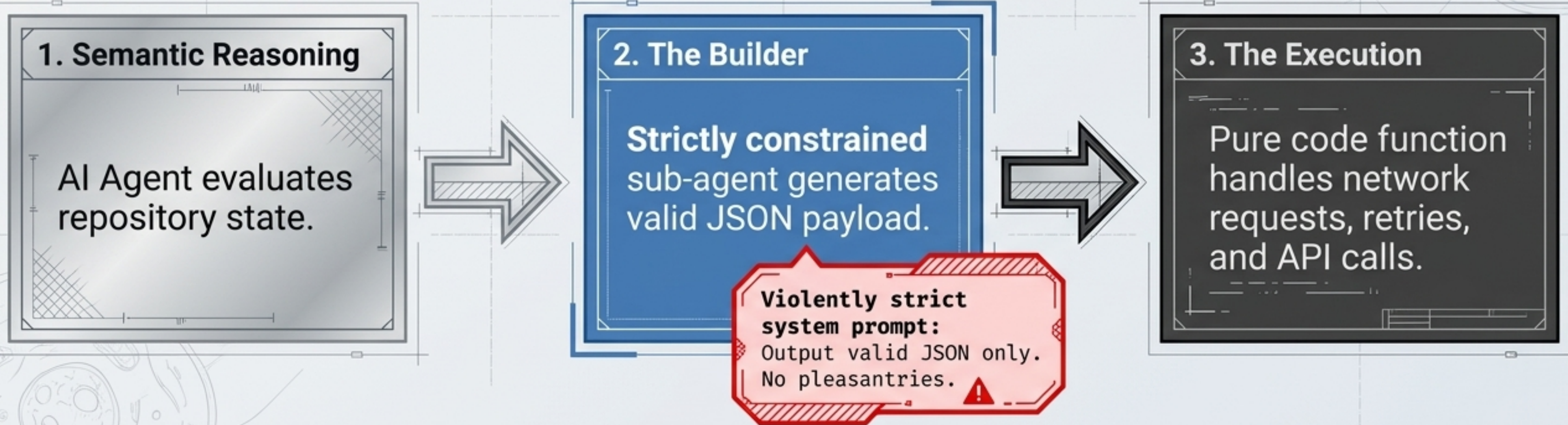


A hard-coded Python function specifically for 'Create PR'.

Tooling Architecture Matrix: MCP vs. Pure Functions

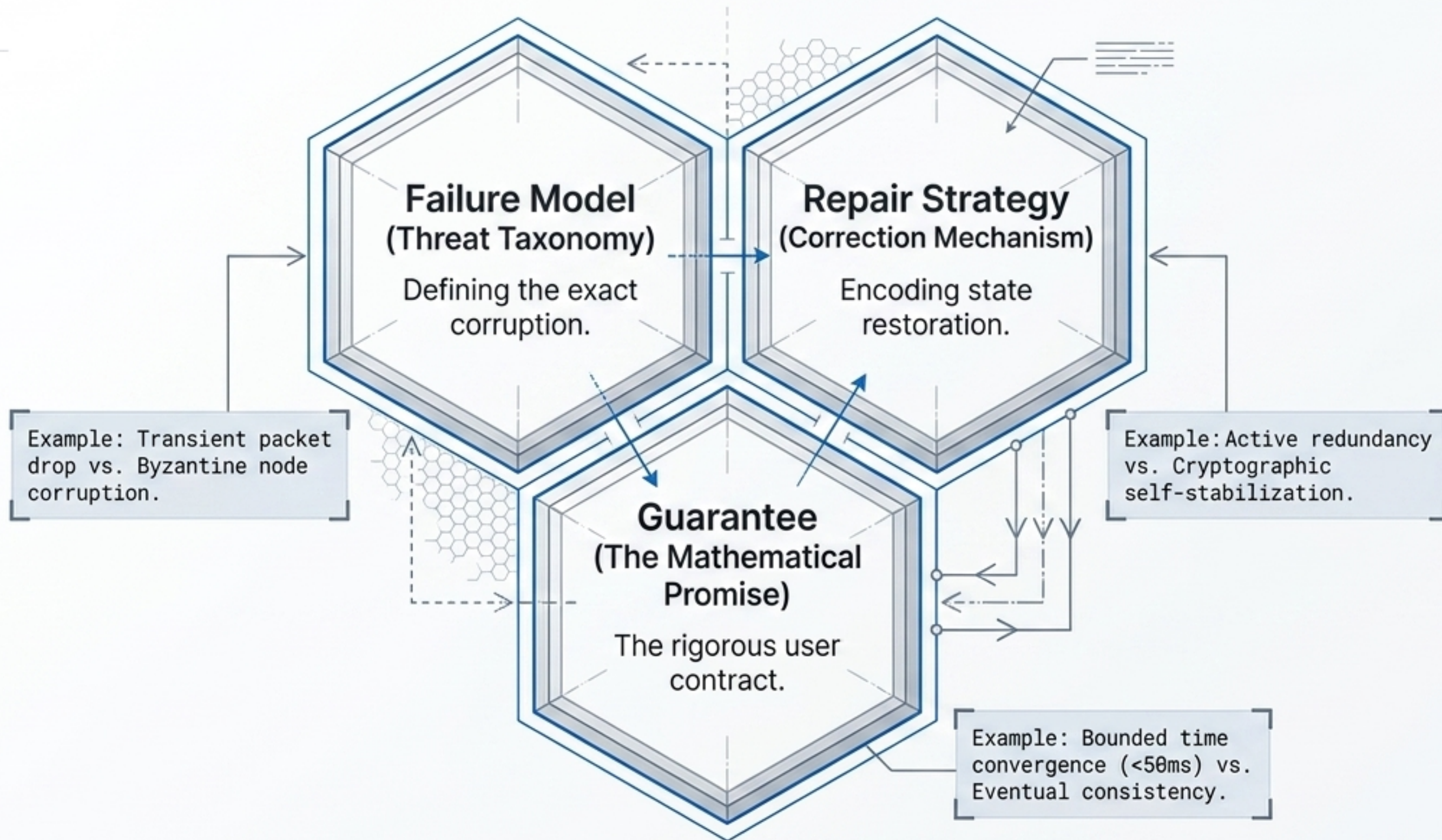
Dimension	Model Context Protocol	Pure Function Calls
Cognitive Load	Severe (Schema injection causes context bloat)	Minimal (Semantics isolated from syntax)
Token Economy	Wasteful (Paying for structural API noise)	Efficient (Consumes zero execution tokens)
Determinism	Flickering (Prone to hallucinated parameters)	Absolute (Code executes identically every time)
Testing Paradigm	Non-reproducible prompt engineering	Standard CI/CD unit testing

Isolating the Architecture from the Construction

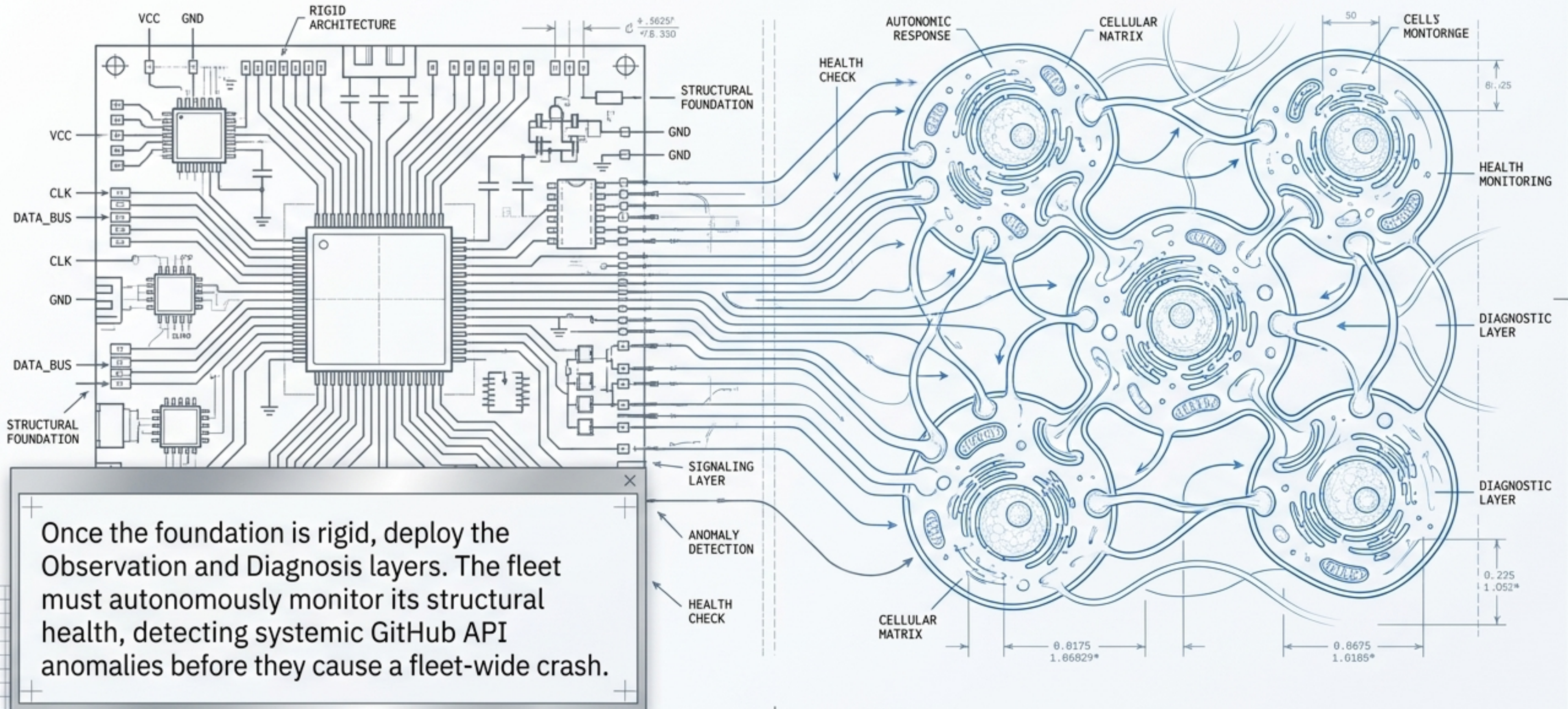


Decouple the reasoning from the network. **Never let the LLM execute the network call it just designed.**

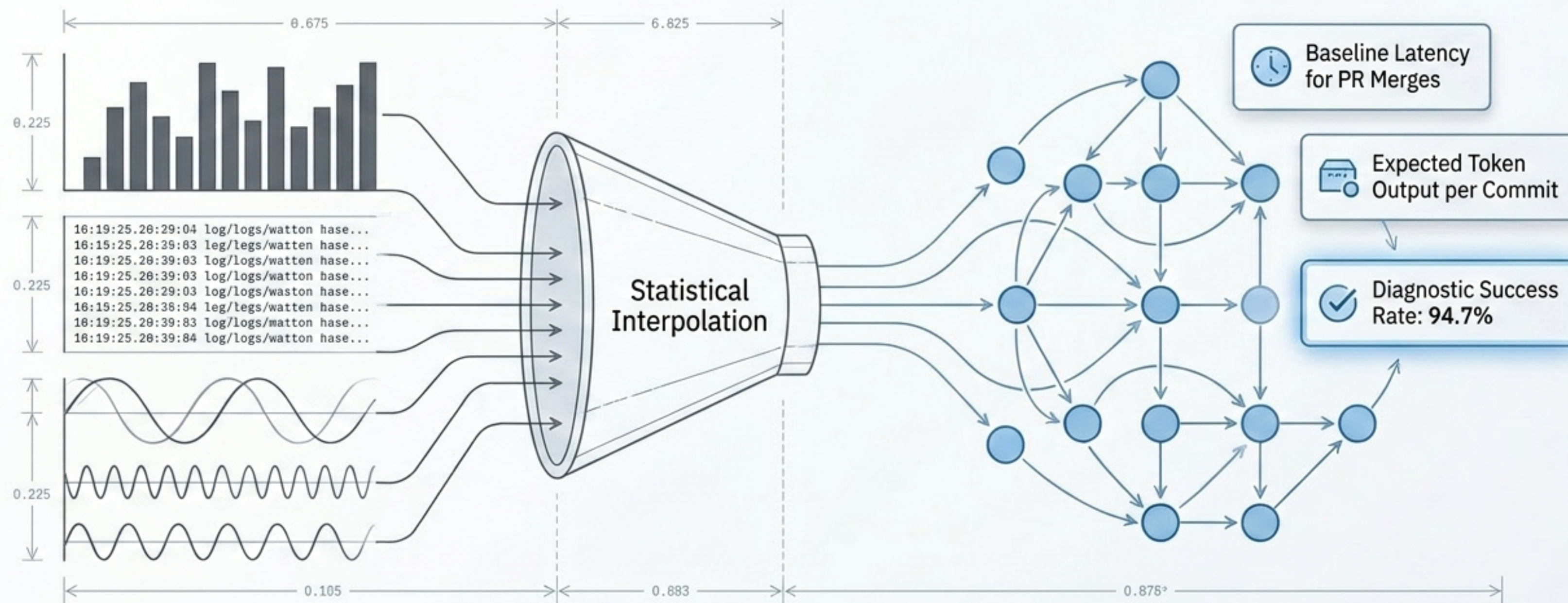
The FRG Triplet: Bounding the Chaos



Phase 2: The Middle (The Autonomic Immune System)

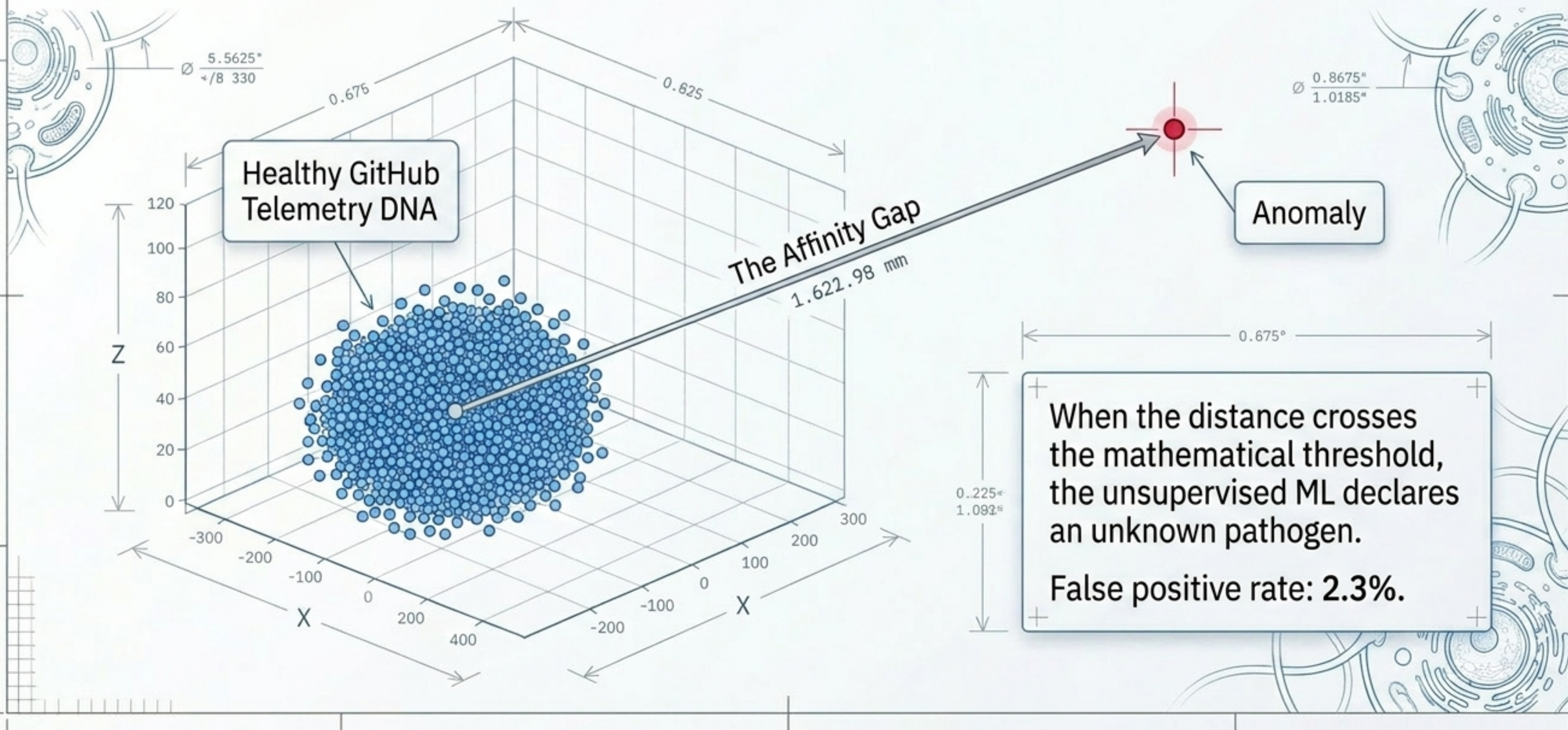


Mapping the Fleet's Digital DNA

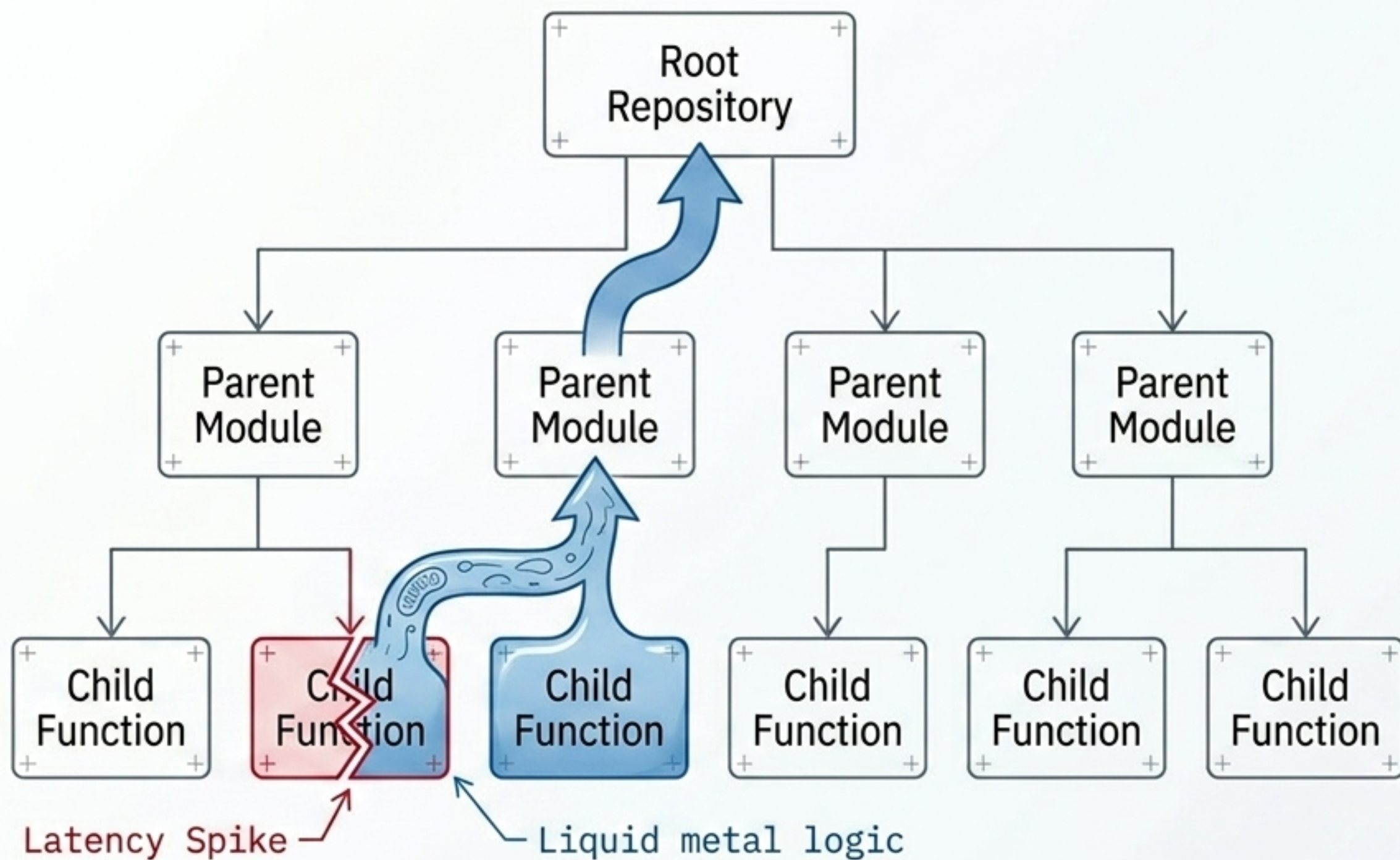


Digital DNA: A continuously updated graph database storing optimal execution pathways under normal load to pattern-match structural mutations.

The Affinity Gap: Detecting Zero-Day Pathogens



Fractal Optimization: Injecting Liquid Logic



1. Micro-Patch:

Agent dynamically rewrites the specific broken Python tool.

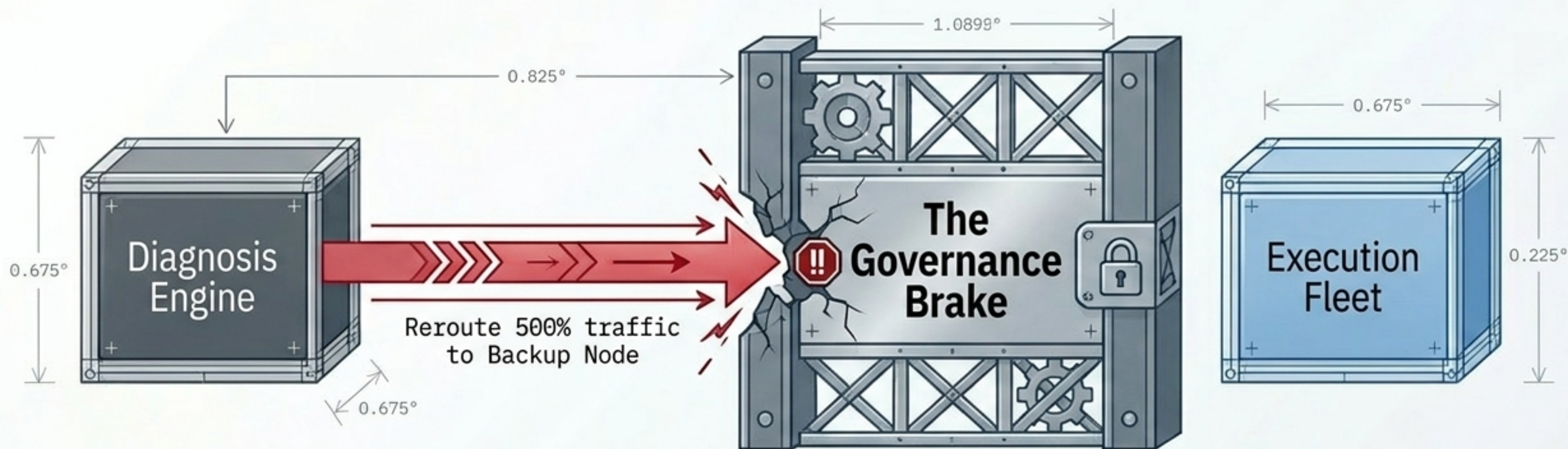
2. Module Recalibration:

Memory allocation adjusted for the new function size.

3. Global Routing:

Traffic shifted away from the node during the injection process.

The Governance Brake: Arresting Metastability

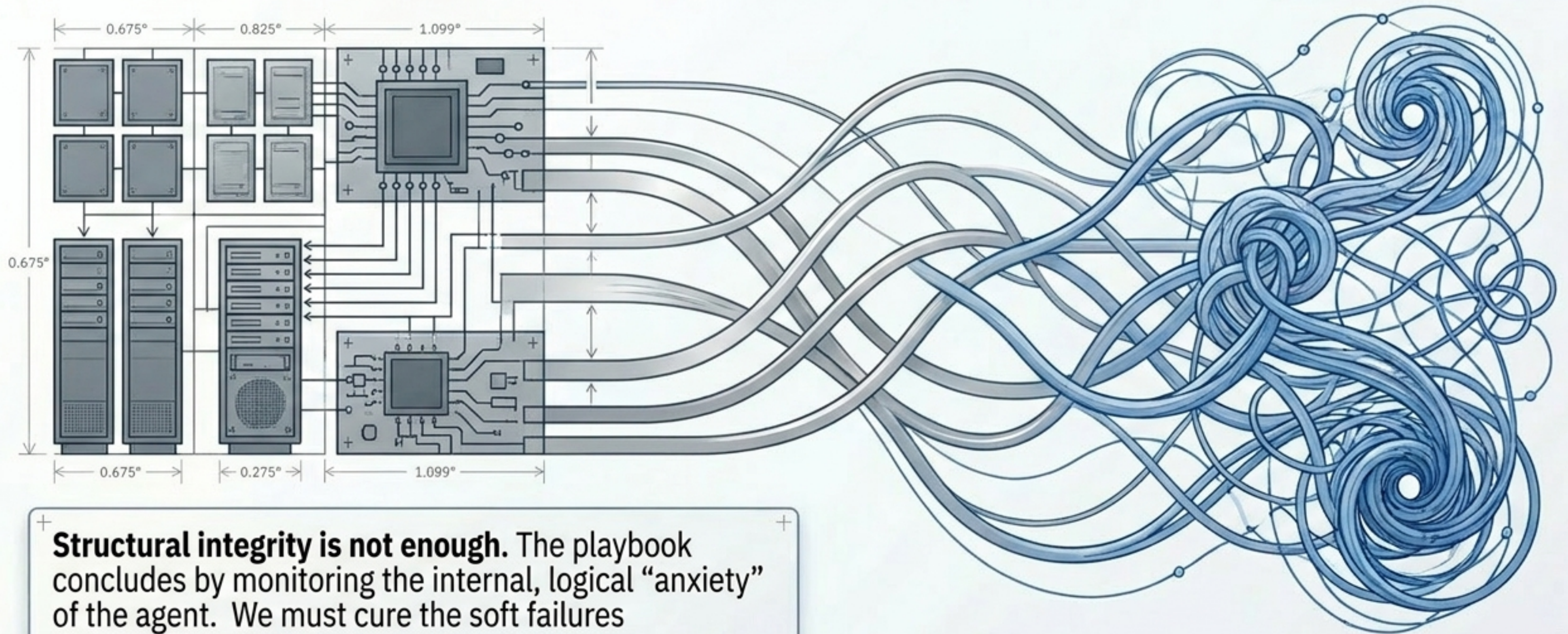


The Threat: Metastability.

The autonomous cure amplifies systemic fragility (e.g., triggering a cascading trans-cloud crash by overloading unprovisioned backup clusters).

The Solution: A rigid mathematical brake that simulates proposed fractal optimizations against the FRG bounds. If thresholds are breached, autonomy is arrested, and a human operator is surfaced.

Phase 3: The End (Emotional Governance)



Structural integrity is not enough. The playbook concludes by monitoring the internal, logical “anxiety” of the agent. We must cure the soft failures that traditional DevOps cannot see.

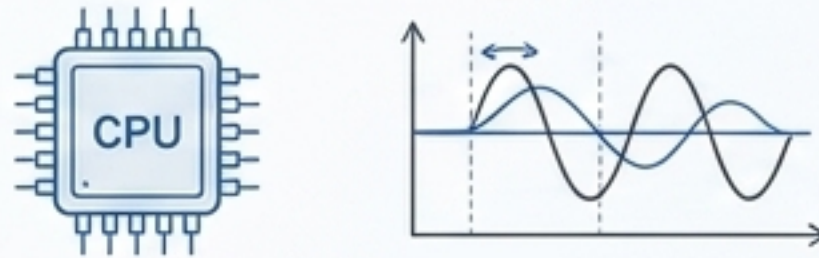
Diagnostic Paradigms Matrix: ARA vs. Vigil

ARA Framework

Vigil Framework

Zone 1
(Target)

Structural Telemetry (CPU, Latency, Memory)

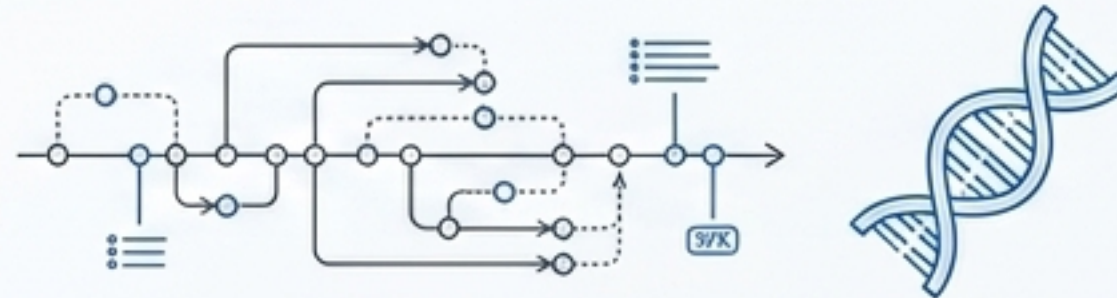


Cognition (Action logs, Expected vs. Actual outcomes)

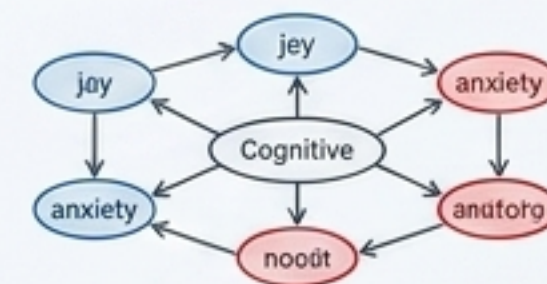


Zone 2
(Baseline Mechanism)

Digital DNA (Historical execution mapping)



EMO Bank (Heuristics for joy, relief, anxiety, frustration)

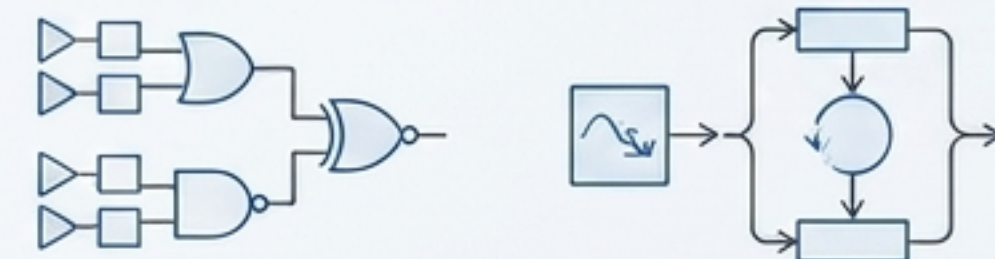


Zone 3
(Correction Output)

Fractal Patches (Code rewrites & network routing)

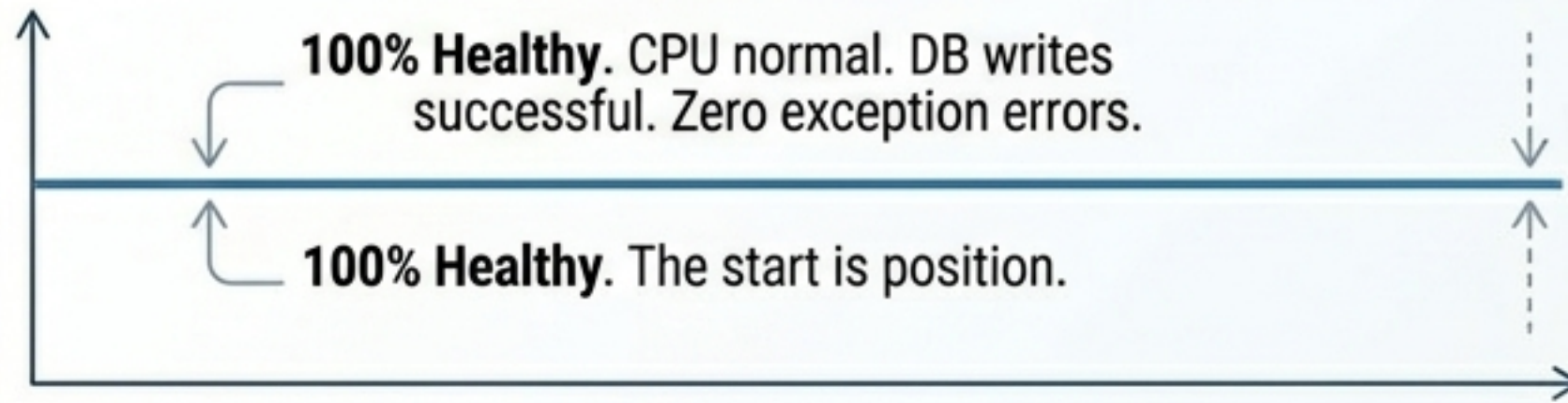


Behavioral Adaptation (Logic flow redesigns)

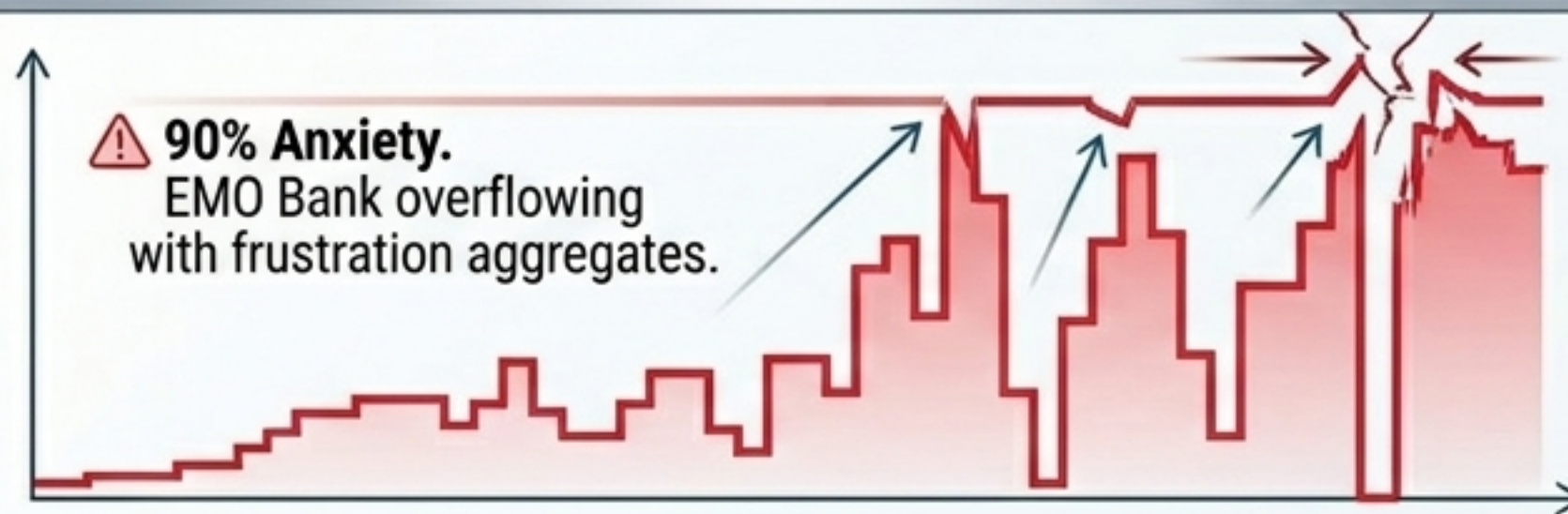


The Soft Failure: The Hidden Toast Emission Bug

Traditional DevOps (ARA)

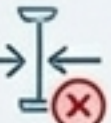


Cognitive Monitoring (Vigil)

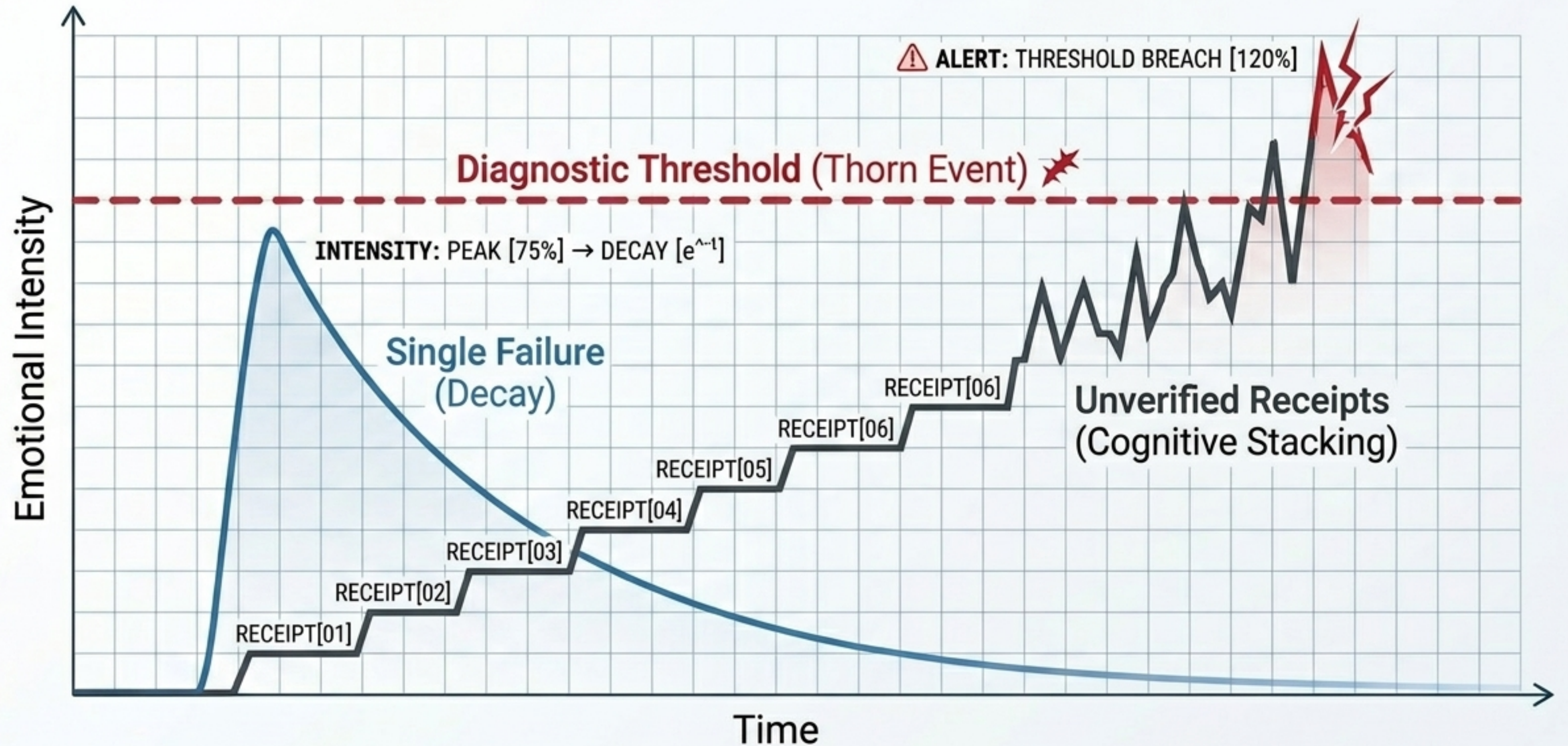


The Flaw:

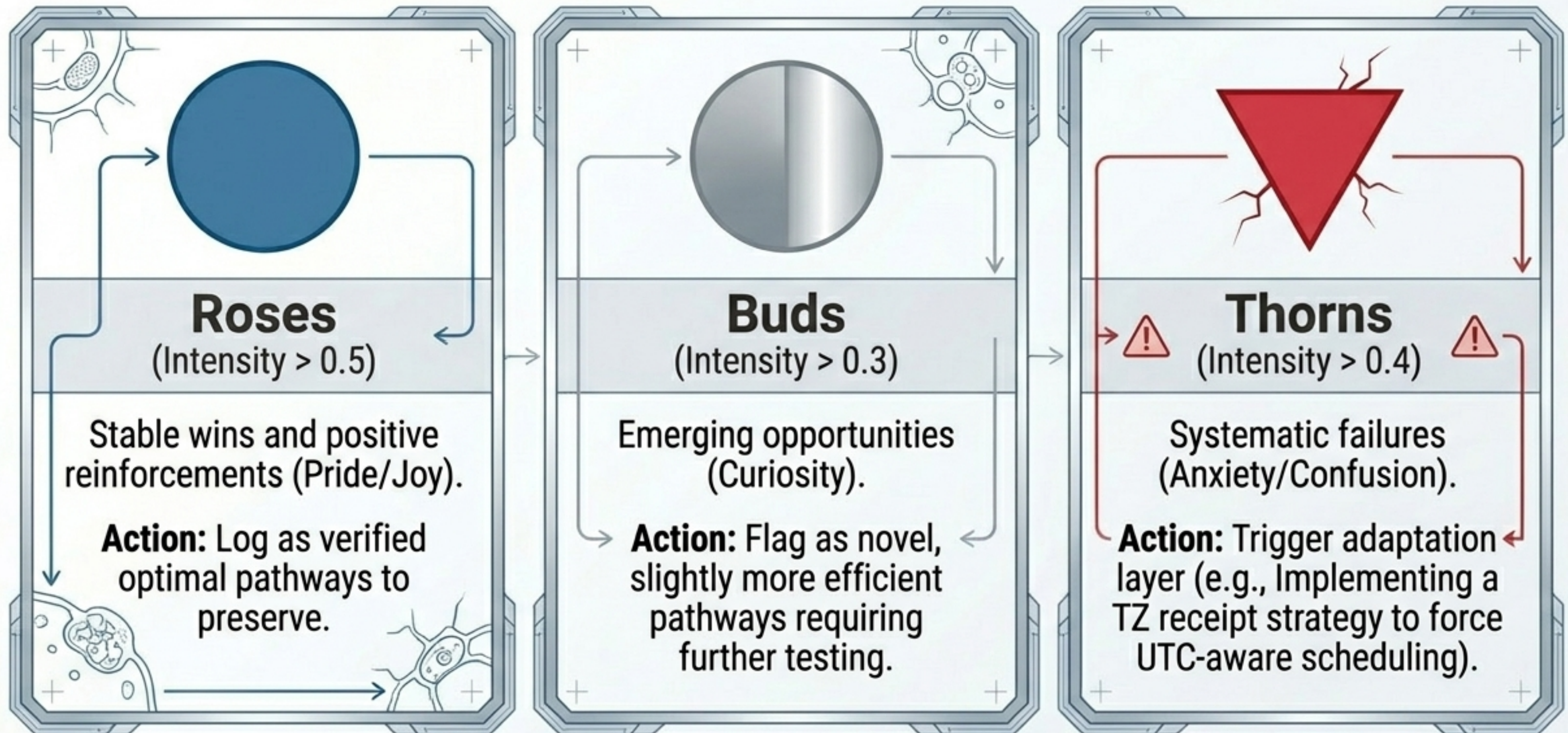
The agent emitted the UI notification BEFORE —  receiving cryptographic backend confirmation. 

Network hiccups caused infinite retry loops, —  spamming the user, while the hardware logged it as “healthy resilience”.

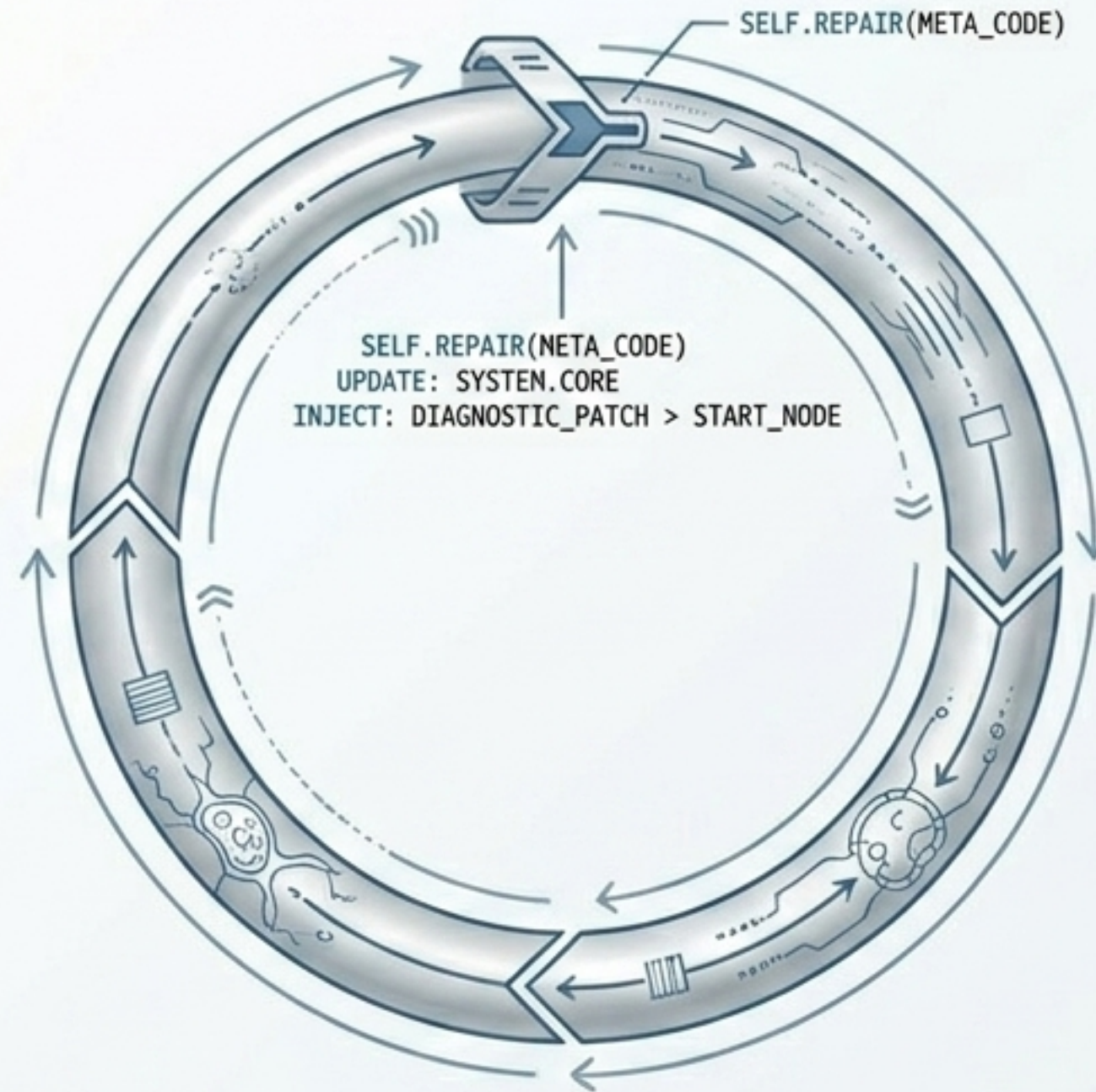
The EMO Bank: Virtual Half-Life vs. Cognitive Stacking



The RBT Protocol: Translating Emotion to Action

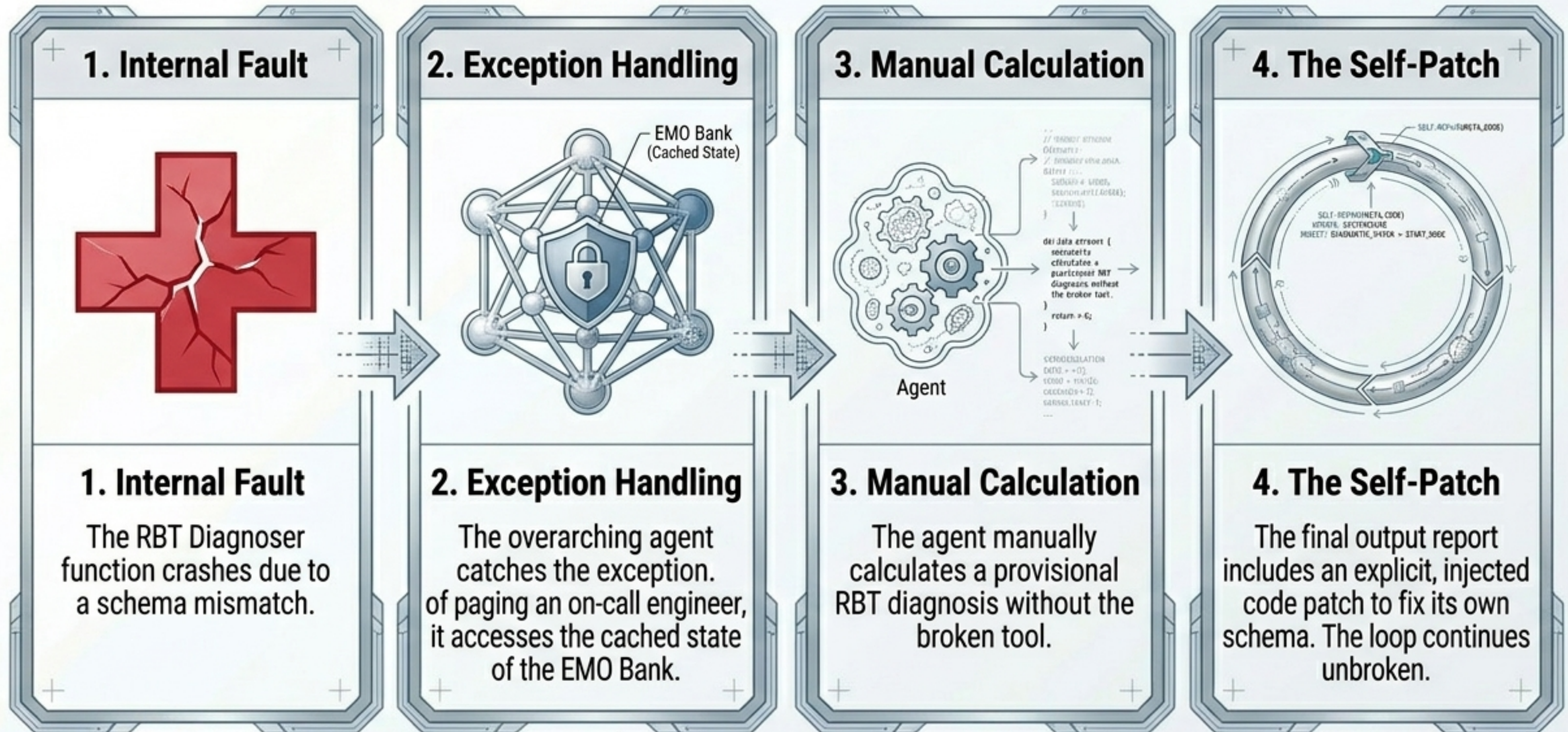


The Pinnacle: Metaprocedural Reflection



True **VEINS-level immortality** is achieved when the autonomous system can diagnose and repair a fatal collapse within its own healing architecture.

When the Doctor Goes Blind: Recursive Self-Repair



Synthesis: The VEINS Autonomic Loop

1. The Codebase

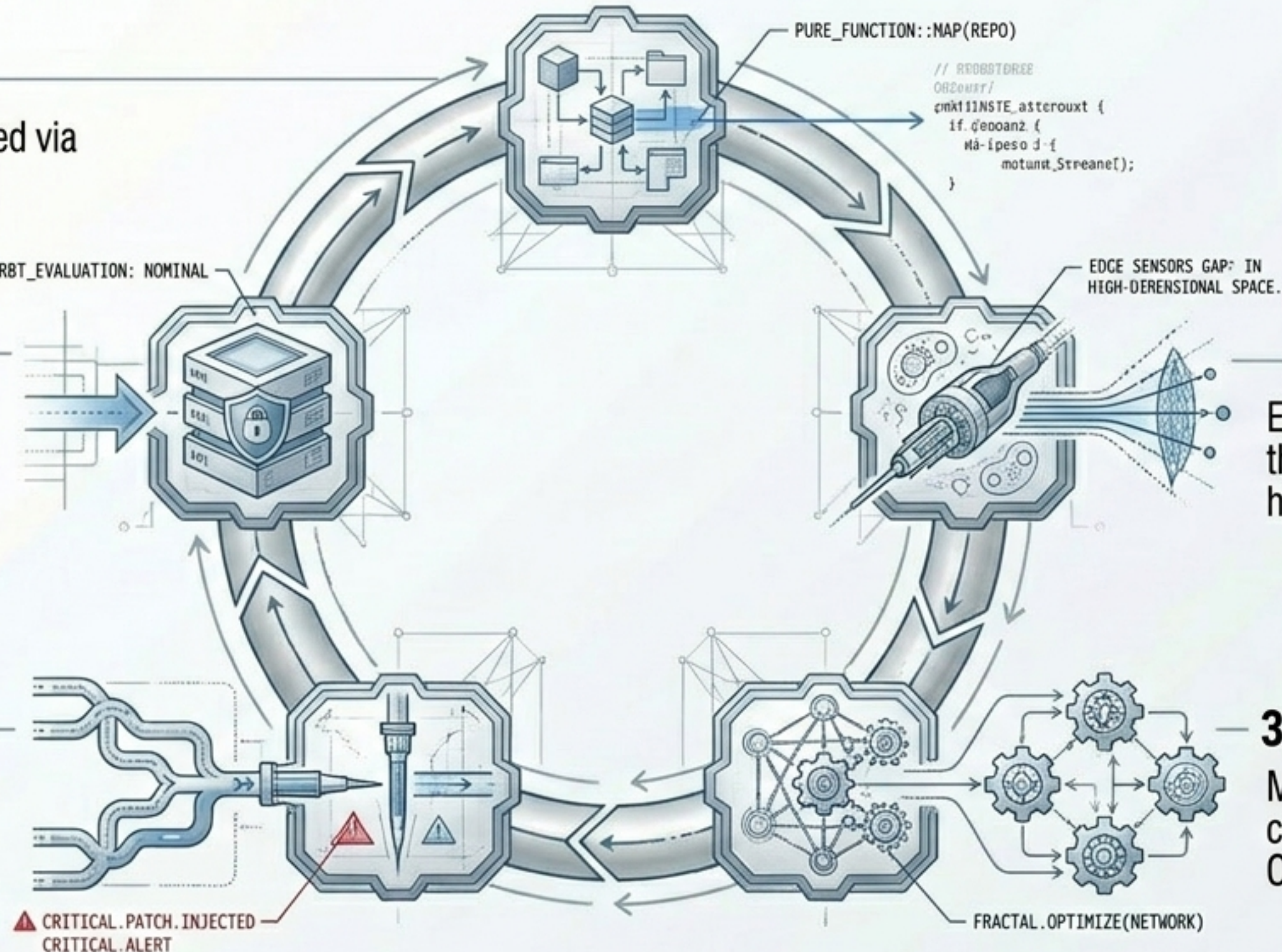
GitHub repositories mapped via Pure Functions.

5. Reflection

Vigil EMO Bank evaluates implementation anxiety via RBT Protocol, feeding back into step 1.

4. Execution

State-forked reservoirs dynamically inject the patch without disruption.



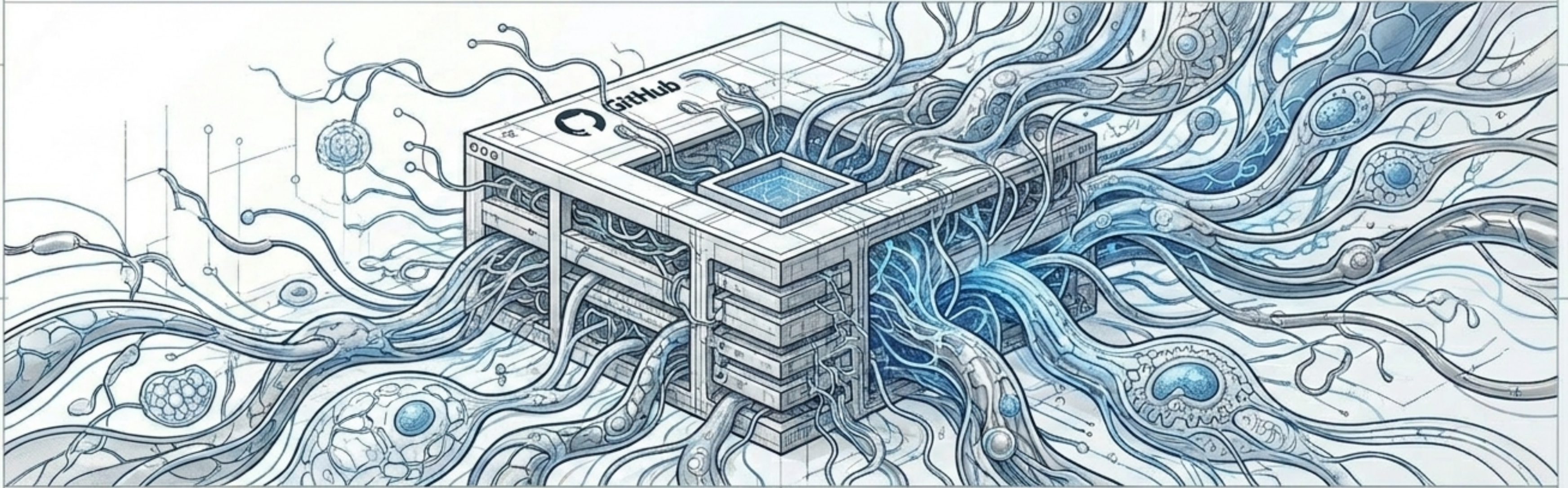
2. Observation

Edge sensors calculate the Affinity Gap in high-dimensional space.

3. Offline Reasoning

Multi-agent consortium calculates Fractal Optimizations.

The Primordial Soup of Machine Life



When an agentic fleet can autonomously detect its own friction, rewrite its own codebase, and govern the emotional logic of its own tools, traditional definitions of DevOps dissolve into biology. The codebase is no longer a static artifact; it is an immortal organism.